🔧 PYINSTALLER FIX - STATIC/TEMPLATES/CONFIG

❌ VẤN ĐỀ:

Khi build bằng PyInstaller, thiếu các files: - `static/` folder - `templates/` folder - `config.key` file

✅ GIẢI PHÁP:

1. Sửa `build.spec` - Thêm `config.key`

```
datas=[
    ('templates', 'templates'),
    ('static', 'static'),
    ('config.json', '.'),
    ('config.key', '.'),  # ← Thêm dòng này
    ('README.md', '.'),
],
```

2. Sửa `app.py` - Handle PyInstaller paths

A. Thêm hàm `get_base_path()`:

```
def get_base_path():
    """Get base path - works for both dev and PyInstaller"""
    if getattr(sys, 'frozen', False):
        # Running in PyInstaller bundle
        return sys._MEIPASS
    else:
        # Running in normal Python
        return os.path.dirname(os.path.abspath(__file__))
```

B. Sử dụng đúng paths:

```
BASE_DIR = get_base_path()
```

```
# Data directory cho config và videos (writable)
if getattr(sys, 'frozen', False):
    # PyInstaller: Dùng user's app data directory
    DATA_DIR = os.path.join(os.path.expanduser('~'), 'EcoHub_QR_Scanner')
else:
    # Dev: Dùng project directory
    DATA_DIR = os.path.dirname(os.path.abspath(__file__))

VIDEOS_DIR = os.path.join(DATA_DIR, "videos")
CONFIG_FILE = os.path.join(DATA_DIR, "config.json")
CONFIG_KEY_FILE = os.path.join(DATA_DIR, "config.key")
```

C. Copy default configs khi first run:

```
# Copy default config files if not exist (for first run)
import shutil
if not os.path.exists(CONFIG_FILE):
    default_config = os.path.join(BASE_DIR, "config.json")
    if os.path.exists(default_config):
        shutil.copy(default_config, CONFIG_FILE)
        print(f"[STARTUP] Copied default config.json to {CONFIG_FILE}")

if not os.path.exists(CONFIG_KEY_FILE):
    default_key = os.path.join(BASE_DIR, "config.key")
    if os.path.exists(default_key):
        shutil.copy(default_key, CONFIG_KEY_FILE)
        print(f"[STARTUP] Copied config.key to {CONFIG_KEY_FILE}")
```

D. Khởi tạo Flask với explicit paths:

```
# Flask app với paths cho PyInstaller
template_folder = os.path.join(BASE_DIR, 'templates')
static_folder = os.path.join(BASE_DIR, 'static')

app = Flask(__name__,
            template_folder=template_folder,
            static_folder=static_folder)

print(f"[STARTUP] BASE_DIR: {BASE_DIR}")
print(f"[STARTUP] Templates: {template_folder}")
print(f"[STARTUP] Static: {static_folder}")
print(f"[STARTUP] Videos: {VIDEOS_DIR}")
```

E. Sử dụng CONFIG_KEY_FILE cho encryption:

```
# Encryption cho sensitive data (use CONFIG_KEY_FILE from DATA_DIR)
encryptor = get_encryptor(CONFIG_KEY_FILE)
```

📁 CẤU TRÚC PATHS:

Khi DEV (python app.py):

```
eco_hub_demo/
├── app.py                  ← Chạy từ đây
├── static/                 ← BASE_DIR = đây
├── templates/              ← BASE_DIR = đây
├── config.json             ← DATA_DIR = BASE_DIR
├── config.key              ← DATA_DIR = BASE_DIR
└── videos/                 ← DATA_DIR = BASE_DIR
```

Khi PYINSTALLER (EcoHub_QR_Scanner.exe):

```
dist/EcoHub_QR_Scanner/
├── EcoHub_QR_Scanner.exe
└── _internal/
    ├── static/             ← BASE_DIR = _MEIPASS (_internal)
    ├── templates/          ← BASE_DIR = _MEIPASS (_internal)
    ├── config.json         ← BASE_DIR = _MEIPASS (read-only)
    └── config.key          ← BASE_DIR = _MEIPASS (read-only)

C:\Users\<user>\EcoHub_QR_Scanner/
├── config.json             ← DATA_DIR (writable copy)
├── config.key              ← DATA_DIR (writable copy)
└── videos/                 ← DATA_DIR (writable)
```

Logic: - **Templates/Static**: Đọc từ `_MEIPASS` (built-in) - **Config files**: Copy từ `_MEIPASS` → User home (first run) - **Videos**: Lưu vào User home (writable)

📋 KIỂM TRA KẾT QUẢ:

Test 1: Files trong dist

```
dir "dist\EcoHub_QR_Scanner\_internal" | findstr /I "static templates config"
```

Kết quả:

```
✅ d----- static/
✅ d----- templates/
```

```
✅ -a---- config.json (899 bytes)
✅ -a---- config.key (44 bytes)
```

Test 2: Chạy EXE

```
dist\EcoHub_QR_Scanner\EcoHub_QR_Scanner.exe
```

Console output:

```
[STARTUP] BASE_DIR: C:\Users\...\AppData\Local\Temp\_MEI123456
[STARTUP] Templates: ...\AppData\Local\Temp\_MEI123456\templates
[STARTUP] Static: ...\AppData\Local\Temp\_MEI123456\static
[STARTUP] Videos: C:\Users\<user>\EcoHub_QR_Scanner\videos
[STARTUP] Copied default config.json to C:\Users\<user>\EcoHub_QR_Scanner\config.json
[STARTUP] Copied config.key to C:\Users\<user>\EcoHub_QR_Scanner\config.key
```

⚙️ BUILD LỆNH:

```
# Clean
powershell -Command "Remove-Item -Recurse -Force 'dist','build'"

# Build
pyinstaller build.spec --clean

# Check
dir "dist\EcoHub_QR_Scanner\_internal" | findstr /I "static templates config"
```

🎯 KẾT QUẢ:

✅ Tất cả files có mặt trong build ✅ Flask tìm thấy templates/static ✅ Config files được copy vào user directory ✅ App chạy OK cả dev và PyInstaller

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📝 FILES ĐÃ SỬA:

1. `build.spec` - Thêm `config.key` vào `datas`

2. `app.py` - Thêm:
3. `get_base_path()` function
4. DATA_DIR for writable files
5. Copy default configs on first run
6. Flask explicit paths (template_folder, static_folder)
7. Use CONFIG_KEY_FILE for encryption